Total No. of Questions : 9 | | Total No. of Printed Pages : 4

Paper Code : 21304

F - 404

B. C. A. (First Semester) Examination, 2022-23

(New Course)

Paper No. BCA-104-N

MATHEMATICS-I

Time : Three Hours] *[Maximum Marks : 70*

Note : Attempt any **five** questions. **All** questions carry equal marks. Symbols used have their usual meanings.

1. (a) Show that the function f defined as:

$$f(x)\begin{cases} 0 & ; \quad x = 0 \\ \frac{1}{2} - x & ; 0 < x < \frac{1}{2} \\ \frac{1}{2} & ; \quad x = \frac{1}{2} \\ \frac{3}{2} - x & ; \frac{1}{2} < x < 1 \\ 1 & ; \quad x = 1 \end{cases}$$

has 3 points of discontinuity and also find them.

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(b) Find the whole area of the curve $a^2y^2=x^3(2a-x)$

2. (a) If $u = \tan^{-1}\frac{xy}{\sqrt{1+x^2+y^2}}$ show that : $\frac{\partial^2 u}{\partial x \partial y} = \frac{1}{(1+x^2+y^2)^{3/2}}$

(b) State and prove jacobian theorem.

3. (a) Evaluate : $\int_0^{2a} \frac{x^{a/2}}{\sqrt{2a-x}}dx$

(b) Prove that : $\nabla^2(r^n\vec{r}) = n(n+3)r^{n-2}\vec{r}$

4. (a) Evaluate : $\lim_{x\to 0} \frac{x^{1/2}\tan x}{(e^x-1)^{1/2}}$

(b) Solve : $(D^2+Dy=8x^2e^{2x} \sin 2x$

5. (a) Find the eccentricity and co-ordinate of the foci of the curve $3x^2-y^2=4$

(b) Find the center, vertices, foci, eccentricity and latus rectum of the hyperbola $9x^2-16y^2-18x-64y-199 = 0$

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6. (a) Find nth derivative of $\tan^{-1}\frac{x}{a}$ with respect to x.

(b) If $I_n = \int_0^{\pi/3} \tan^n x\,dx$, show that

$(n-1)(I_n + I_{n-2}) = (\sqrt{3})^{n-1}$

7. (a) Show that the vectors $\bar{a} = 3\hat{i} - 2\hat{j} + \hat{k}, \hat{b} = \hat{i} - 3\hat{j} + 5\hat{k}$ and $\hat{c} = 2\hat{i} + \hat{j} - 4\hat{k}$ form a right angled triangle.

(b) Obtain the reduction formula for

$\int \sec^n x\,dx$

8. (a) If $x = r\sin\theta\cos\phi$

$y = r\sin\theta\sin\phi$

$z = r\cos\theta$

Show that : $\frac{\partial(x,y,z)}{\partial(r,\theta\phi)} = r^2\sin\theta$

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(b) Determine the value of a,b,c for which the function:

$$f(x) = \begin{cases} \frac{\sin(a+1)x + \sin x}{x}; & x < 0 \\ c; & x = 0 \\ \frac{(x+bx^2)^{1/2} - x^{1/2}}{bx^{3/2}}; & x > 0 \end{cases}$$

is continuous at x=0

9. (a) Show that the volume of a sphere of radius 'a' is $\frac{4}{3}\pi a^3$

(b) solve : $(D^4+m^4)y = 0$

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